

Practice Lab 11

Nested Queries and Joins

1. Display the customers' name and phone number along with their current credit which is greater than 15000 by using INNER JOIN.

```
mysql> SELECT
->     c.name, c.phoneNum, l.current_credit
-> FROM
->     customer c
-> INNER JOIN
->     loanaccounts l
-> ON
-> c.customer_id = l.customer_id
-> WHERE
->     l.current_credit > 15000;
```

name	phoneNum	current_credit
Zubair ALi	92343212312	70000
Ahtasham Khan	92343212322	38000
M. Kareem	92323212322	60000
Imran Khan	92313212321	30000

4 rows in set (0.01 sec)

2. Display customer's name and address along with the maximum credit of the customer whose name is Imran Khan.

```
mysql> SELECT
->      c.name, c.Address, l.max_credit
-> FROM
->      customer c
-> INNER JOIN
->      loanaccounts l
-> ON
-> c.customer_id = l.customer_id
-> WHERE
->      c.name = 'Imran Khan';
+-----+-----+-----+
| name      | Address                | max_credit |
+-----+-----+-----+
| Imran Khan | 16-A Johar Town Lahore |      75000 |
+-----+-----+-----+
1 row in set (0.00 sec)

mysql>
```

3. Display the names, phone numbers and current credit of those customers whose current credit is NULL by using LEFT JOIN.

```
mysql> SELECT
->     c.name, c.phoneNum, l.current_credit
-> FROM
->     customer c
-> LEFT JOIN
->     loanaccounts l
-> ON
->     c.customer_id = l.customer_id
-> WHERE
->     current_credit IS NULL;
+-----+-----+-----+
| name          | phoneNum      | current_credit |
+-----+-----+-----+
| Waqar Azeem   | 92353212329   | NULL          |
| Ahsan ALi     | 923126455982  | NULL          |
| Imtiaz Khan   | 92323212323   | NULL          |
| Shafqat Ali   | 92323212326   | NULL          |
+-----+-----+-----+
4 rows in set (0.00 sec)
```

4. Display the complete data of those customers who have applied for loan but not for the saving accounts by using IN and NOT commands.

```
mysql> SELECT
->      *
-> FROM
->      customer
-> WHERE
->      customer_id IN (
->
Display all 826 possibilities? (y or n)
-> SELECT
->          customer_id
->      FROM
->          loanaccounts)
->      AND
->      customer_id NOT IN (
->          SELECT
->              customer_id
->          FROM
->              savingaccounts);
```

customer_id	name	Address	phoneNum
352021832671	Anwar Majeed	58-B Wapda Town,Multan	92343212311
352021832672	Zubair ALi	16-A Johar Town Lahore	92343212312
352021832673	Ahtasham Khan	33-Y Nazamabad, Karachi	92343212322

```
3 rows in set (0.01 sec)
```

5. Display the complete record of the transaction made by the customer along with the customer Name and the Salesman Name

```
mysql> SELECT t.*, c.name AS customer_Name, e.name AS salesman_Name FROM transaction t LEFT JOIN savingaccounts s ON t.savingAccount_number = s.Account_number LEFT JOIN loanaccounts l ON t.loanAccount_number = l.Account_number LEFT JOIN customer c ON c.customer_id = s.customer_id OR c.customer_id = l.customer_id LEFT JOIN employee e ON t.eid = e.eid;
```

transaction_id	transaction_amount	transaction_date	loanAccount_number	savingAccount_number	transactionDetails	eid	customer_Name	salesman_Name
tr101	-30000	2016-01-08	NULL	sbABL303	debited amount for personal use	ABL0101	M. Ali	Haider
tr1010	-26000	2017-05-07	NULL	sbUBL105	For home renovation	UBL0202	Waqar Azeem	Sarmad
tr1011	27000	2017-06-06	NULL	sbMCB101	For Employees salaries	MCB202	Intiaz Khan	Zubair
tr1012	-23000	2018-01-05	NULL	sbMCB101	For purchasing LED	MCB203	Intiaz Khan	Zaryab
tr1013	-32000	2018-02-04	NULL	sbMCB100	debited amount for personal use	MCB203	Shafaqat Ali	Zaryab
tr1014	30000	2017-03-03	NULL	sbMCB100	For purchasing LED	MCB202	Shafaqat Ali	Zubair
tr102	800000	2017-02-07	NULL	sbABL303	credited amount of sold car	ABL0102	M. Ali	Ayesha
tr103	30000	2016-03-06	NULL	sbUBL204	For Office use	UBL0101	Imran Khan	Sara
tr104	28000	2017-04-05	NULL	sbUBL204	For utility bills	UBL0103	Imran Khan	Samreen
tr105	-25000	2016-05-04	NULL	sbUBL103	For Employees salaries	UBL0201	M. Karzem	Ali
tr106	20000	2018-01-03	NULL	sbUBL103	salary credited	UBL0202	M. Karzem	Sarmad
tr107	-15500	2016-02-02	NULL	sbABL107	debited amount for personal use	ABL0203	Ali Ahmad	Kamran
tr108	12000	2016-03-01	NULL	sbABL107	For Office use	ABL0203	Ali Ahmad	Kamran
tr109	14000	2017-04-08	NULL	sbUBL105	For utility bills	UBL0201	Waqar Azeem	Ali
tr201	900000	2017-04-09	lbABL501	NULL	For purchasing AC	ABL0202	Anwar Majeed	Anwar
tr2010	60000	2017-05-02	lbUBL301	NULL	For purchasing Mobile Phone	UBL0103	Imran Khan	Samreen
tr2011	90000	2017-07-03	lbUBL201	NULL	For purchasing AC	UBL0201	M. Karzem	Ali
tr2012	95000	2017-08-04	lbUBL201	NULL	For purchasing Laptop	UBL0202	M. Karzem	Sarmad
tr2013	20000	2017-09-05	lbMCB701	NULL	For purchasing AC	MCB202	Ali Ahmad	Zubair
tr2014	15000	2017-10-08	lbMCB701	NULL	For purchasing Washing Machine	MCB203	Ali Ahmad	Zaryab
tr202	85000	2017-04-10	lbABL501	NULL	For purchasing LED	ABL0203	Anwar Majeed	Kamran
tr203	95000	2017-05-09	lbABL502	NULL	For purchasing Fridge	ABL0301	Zubair Ali	Nouman
tr204	100000	2017-04-11	lbABL502	NULL	For purchasing Furniture for office	ABL0303	Zubair Ali	Tanveer
tr205	175000	2017-03-11	lbABL601	NULL	For purchasing Home Furniture	ABL0203	Ahtasham Khan	Kamran
tr206	150000	2017-04-05	lbABL601	NULL	For purchasing LED	ABL0202	Ahtasham Khan	Anwar
tr207	40000	2017-01-09	lbUBL401	NULL	For purchasing UPS	UBL0201	M. Ali	Ali
tr208	30000	2017-02-09	lbUBL401	NULL	For purchasing LED	UBL0201	M. Ali	Ali
tr209	70000	2017-03-09	lbUBL301	NULL	For purchasing Car tyres	UBL0101	Imran Khan	Sara

28 rows in set (0.00 sec)

6. By using IN command display the complete record of customers and the record of their saving accounts along with the branch id of the bank.

```
mysql> SELECT
-> c.*, s.*, s.branch_id
-> FROM
-> customer c
-> INNER JOIN
-> savingaccounts s
-> ON
-> c.customer_id = s.customer_id
-> WHERE
-> c.customer_id IN (
->
Display all 826 possibilities? (y or n)
-> SELECT
-> customer_id
-> FROM
-> savingaccounts);
```

customer_id	name	Address	phoneNum	Account_number	branch_id	customer_id	current_balance	branch_id
332021832677	Ali Ahmad	90 Kemari Town, Karachi	92323212324	sbABL107	ABL002	332021832677	32800	ABL002
352017898532	Ahsan Ali	19-J Johar Town Lahore	923126455982	sbABL109	ABL001	352017898532	59000	ABL001
352021832674	M. Ali	76-C Faisal Town Lahore	92343212321	sbABL303	ABL001	352021832674	36000	ABL001
362021832676	Shafqat Ali	32-H Liari, Karachi	92323212326	sbMCB100	MCB002	362021832676	32800	MCB002
362021832675	Imtiaz Khan	21-D Johar Town Lahore	92323212323	sbMCB101	MCB002	362021832675	32800	MCB002
332021832676	M. Kareem	78 Korangi Town, Karachi	92323212322	sbUBL103	UBL002	332021832676	32800	UBL002
332021832678	Waqar Azeem	23-C Qasimabad, Haiderabad	92353212329	sbUBL105	UBL002	332021832678	32800	UBL002
332021832675	Imran Khan	16-A Johar Town Lahore	92313212321	sbUBL204	UBL001	332021832675	930000	UBL001

8 rows in set (0.01 sec)

7. Display the name and address of the customers along with their branch address and bank name;

```
mysql> SELECT
->   c.name, c.Address, b.branch_address, bk.bank_name
-> FROM
->   customer c
-> INNER JOIN
->   savingaccounts s
-> ON
->   c.customer_id = s.customer_id
-> INNER JOIN
->   branch b
-> ON
->   s.branch_id = b.branch_id
-> INNER JOIN
->   bank bk
-> ON
->   b.bank_id = bk.bank_id;
+-----+-----+-----+-----+
| name      | Address                | branch_address          | bank_name                |
+-----+-----+-----+-----+
| Shafqat Ali | 32-H Liari, Karachi   | 15, Jail road branch Lahore | Muslim Commercial Bank Limited |
| Imtiaz Khan | 21-D Johar Town Lahore | 15, Jail road branch Lahore | Muslim Commercial Bank Limited |
| Ali Ahmad   | 90 Kemari Town, Karachi | Johar Town branch Lahore | Allied Bank Limited        |
| Ahsan Ali   | 19-J Johar Town Lahore | Iqbal Town branch Lahore | Allied Bank Limited        |
| M. Ali      | 76-C Faisal Town Lahore | Iqbal Town branch Lahore | Allied Bank Limited        |
| M. Kareem   | 78 Korangi Town, Karachi | Thokar niaz baig branch Lahore | United Bank Limited        |
| Waqar Azeem | 23-C Qasimabad, Haidarabad | Thokar niaz baig branch Lahore | United Bank Limited        |
| Imran Khan  | 16-A Johar Town Lahore | Baghban pura branch Lahore | United Bank Limited        |
+-----+-----+-----+-----+
8 rows in set (0.00 sec)
```


8. Display the branch id and branch address of the Allied Bank Limited by using IN command.

```
mysql> SELECT
->     branch_id, branch_address
-> FROM
->     branch
-> WHERE
->     bank_id IN (
->
Display all 826 possibilities? (y or n)
-> SELECT
->     bank_id
-> FROM
->     bank
-> WHERE
->     bank_name = 'Allied Bank Limited');
+-----+-----+
| branch_id | branch_address |
+-----+-----+
| ABL001    | Iqbal Town branch Lahore |
| ABL002    | Johar Town branch Lahore |
| ABL003    | Barkat Market branch Lahore |
+-----+-----+
3 rows in set (0.00 sec)

mysql> |
```

9. Display the names, designations of the employees along with their branch address and bank name by using multiple RIGHT JOINS.

```
mysql> SELECT
->   e.name, e.designation, b.bank_id, br.branch_address
-> FROM
->   employee e
-> RIGHT JOIN
->   branch br
-> ON
->   e.branch_id = br.branch_id
-> RIGHT JOIN
->   bank b
-> ON
->   br.bank_id = b.bank_id;
+-----+-----+-----+-----+
| name | designation | bank_id | branch_address |
+-----+-----+-----+-----+
| Aslam | Cashier | 1 | Moon Market branch Lahore |
| Ahmad | Manager | 1 | Moon Market branch Lahore |
| Zubair | Manager | 1 | 15, Jail road branch Lahore |
| Zaryab | Cashier | 1 | 15, Jail road branch Lahore |
| Haider | Cashier | 2 | Iqbal Town branch Lahore |
| Ayesha | Manager | 2 | Iqbal Town branch Lahore |
| Anwar | Cashier | 2 | Johar Town branch Lahore |
| Kamran | Manager | 2 | Johar Town branch Lahore |
| Nouman | Manager | 2 | Barkat Market branch Lahore |
| Tanveer | Cashier | 2 | Barkat Market branch Lahore |
| Sara | Manager | 3 | Baghban pura branch Lahore |
| Samreen | Cashier | 3 | Baghban pura branch Lahore |
| Ali | Manager | 3 | Thokar niaz baig branch Lahore |
| Sarmad | Cashier | 3 | Thokar niaz baig branch Lahore |
+-----+-----+-----+-----+
14 rows in set (0.01 sec)
```

10. Display the branch id of those branches which have assigned loan accounts and saving accounts to the customers by using multiple IN commands.

```
mysql> SELECT
->     branch_id
-> FROM
->     branch
-> WHERE
->     branch_id IN ( SELECT
->                     branch_id
->                     FROM
->                         loanaccounts)
-> AND branch_id IN (SELECT
->                     branch_id
->                     FROM
->                         savingaccounts);
+-----+
| branch_id |
+-----+
| MCB002    |
| ABL002    |
| UBL001    |
| UBL002    |
+-----+
4 rows in set (0.00 sec)
```